Question 1

Robin Blake is a research analyst for Star Investments and covers the basic chemicals industry. After carrying out a thorough research backed by fundamental analysis, she has concluded that the stock of Petrovas is overpriced at its current traded level compared to its intrinsic value. Petrovas is an important investment banking client of the firm, and has recently invited bids from various investment banks including Star Investments to underwrite a debt issue. Blake's supervisor has asked her to complete her research report on Petrovas, as soon as possible. Blake is worried that if her recommendation is negative, it will create a rift between the client and her firm and Star might not be awarded the contract. By not issuing the correct recommendation, Blake most likely violates the CFA Institute Standard relating to:

A. diligence and reasonable basis.
B. independence and objectivity.
C. knowledge of the law.

B is correct. Blake does not violate Standard V (A) – Diligence and Reasonable Basis, since her conclusion was determined after a thorough analysis. She does not violate Standard I (A) – Knowledge of the Law. She violates Standard I (B) – Independence and Objectivity, by not issuing an unfavorable rating of Petrovas on account of the investment banking relationship between Star and Petrovas. Guidance for Standards I-VII

Question 2

Rana Fahim is a fixed-income analyst and portfolio manager for Tameed Investments Company (TMC), a small firm specializing in fixed-income investments. A former colleague of Fahim who works for a credit-rating agency informs her on June 20 that one of the bonds that the company has under review will undergo a rating change. The same bond is in two of Fahim's portfolios. Fahim, however, waits for the rating change to be announced through both print and electronic media on June 22 by the credit-rating agency. But in the afternoon of June 21, the information is broadcasted country-wide through a well-reputed financial news service. Fahim decides to immediately trade the bonds in her two portfolios instead of waiting for the agency's announcement. Does Fahim violate the CFA Institute Standards by trading on the financial news ahead of the agency's announcement on June 22?

A. **No**.
B. Yes, relating to material nonpublic information.
C. Yes, relating to market manipulation.

A is correct. Fahim did not act on the material nonpublic information she possessed but waited until it was made public through the financial news service. Guidance for Standards I-VII

Question 3

Steven Andrade, CFA, has consistently posted above-market returns for the assets he has managed. Based on his reputation, he is being considered by a wealthy family to manage their portfolio. The family lays out their return objectives and constraints to Andrade. They have certain conditions over what stocks cannot be a part of their portfolio such as companies involved in the production or sale of meat processing and packaging, alcohol, and genetically modified seeds. Andrade expresses his concern that these constraints will affect his investment strategy and performance. To substantiate, he cites the impact of excluding Monsanto as an example. Eventually, he respects the family's beliefs and agrees to manage their portfolio with the constraints. Has Andrade most likely violated the CFA Institute Standards of Professional Conduct?

A. Yes, because he was considered based on his past performance and investment strategy.
B. No.
C. Yes, because the constraints will limit the returns on their portfolio.

B is correct. His actions are in accordance with Standard III (A) Loyalty, Prudence, and Care. Andrade must respect the wishes of the client and design a strategy that best suits their return and risk profile. Guidance for Standards I-VII

Question 4

Ruby Stern, an analyst with Salzar Asset Management Company, is currently recommending the purchase of PPK, a paper and packaging company. Stern's well analyzed and thoroughly researched report has been published which substantiates the recommendation. Few days after publishing her report, Stern receives a call from the client-relationship office of PPK, inviting her to a tour of PPK's facility and a chance to speak with the CFO of the firm, Greg Vaughlis. Stern declines the travel arrangements offered by PPK, but after receiving written approval from her supervisor, and the chief investment officer (CIO), visits the facility and is later invited for coffee with Vaughlis at a nearby expensive restaurant only frequented by the firm's top management. Upon her return Stern provides complete details of her visit to the factory and meeting with Vaughlis to her supervisor and the CIO through a memo, but does not mention about the restaurant. Stern most likely violated the CFA Institute Standard relating to:

A. Additional compensation arrangements.
B. Loyalty.
C. Misrepresentation.

A is correct. Stern has not violated CFA Standard I (C) – Misrepresentation. Members and candidates must not knowingly make any misrepresentations relating to investment analysis, recommendations, actions, or other professional activities. She has not violated Standard IV (A) – Loyalty, according to which she must act for the benefit of her employer, and not deprive them of the advantage of her skills, abilities, divulge confidential information, or otherwise cause harm to them. Stern could possibly be in violation of IV (B) – Additional Compensation Arrangement since Stern did not notify her company of the meeting in the restaurant. Guidance for Standards I-VII

Question 5

Sarah Ryan, CFA, worked as a wealth manager for a large full-service financial services firm for 10 years. She leaves the firm to start her independent financial planning business, Ryan Planning Services. Leigh Smith, one of Ryan's former clients, e-mails her with a query on a mid-cap fund Ryan had recommended for Smith's portfolio in the last report she wrote at her previous firm. Ryan responds to Smith's query, and solicits her to become a client of Ryan Planning Services. Ryan also e-mails her personal clients a copy of the planning reports she worked on in the last 4 weeks before leaving the firm. This was done without her former employer's knowledge. Which of the following actions by Ryan most likely violates the CFA Institute Standards of Professional Conduct?

A. No written approval from employer to email planning reports to her personal clients.
B. Soliciting Smith to sign up with Ryan Planning Services.
C. Starting an independent financial planning business as it is a conflict of interest.

A is correct. Ryan violated Standard V (C) – Record Retention by emailing reports sent to clients without obtaining written consent from the firm. B and C are incorrect because there is no evidence of violation of Standard IV (A) – Loyalty to Employer that shows Ryan carried client lists with her or that she did not obtain permission to start independent practice. Guidance for Standards I-VII

Question 6

Deborah Pope, CFA, is an analyst covering the financial sector at FCI Investments. She is attending a banking conference in Orlando. Over lunch, she overhears the President of Suez Bank tell his CFO that KF Airlines, one of the bank's largest clients, is in financial trouble. He also adds that this information is not yet public. If KF Airlines starts defaulting on payments, the asset quality of Suez Bank will deteriorate, and it is highly unlikely that the Bank will meet its earnings forecasts. Pope had issued a buy recommendation on Suez Bank last month. She calls her manager at FCI to convey the information she has just learnt on Suez and KF. Pope's manager asks her to revise the report to sell. To comply with the CFA Institute Standards of Professional Conduct, Pope should most likely:

A. revise the report.
B. act on the information as it is in the best interest of her clients.
C. encourage the President of Suez to make the information on KF Airlines public.

C is correct. The information on KF Airlines mentioned by the President of Suez Bank is material nonpublic information. Pope should encourage the President to make it public. Guidance for Standards I-VII

Question 7

Thomas Reilley works for Galaxy Asset Management (GAM) as a portfolio manager. He meets with a high-net-worth client who would like to invest both in government bonds and growth stocks. Reilley discusses the client's expectation of risk and return, constraints, and various performance measurement benchmarks. He realizes that a balanced fund would be the appropriate investment strategy for this particular client. Hence, Reilly makes the following two statements to his client regarding investments in a long-term government bonds, fixed-income fund and a growth stock fund:

I. "The payment of the bonds is guaranteed by the government; therefore, the default risk is negligible."

II. "An investment in our growth stock fund will earn an average 12% rate of return per annum based on the historical market performance."

Did Reilley's statements violate the CFA Institute Code and Standards in the above statements?

A. Both statements I and II.
B. Only statement II.
C. Only statement I.

B is correct. Only statement II violated the code relating to Standard I (C) – Misrepresentation. Statement I is a factual statement that discloses accurate information about the investment instrument. Statement II guarantees a specific rate of return for a mutual fund, which is an opinion stated as a fact, therefore violates Standard I (C). Guidance for Standards I-VII

Question 8

Arceus Reed, CFA, manages a portfolio of small cap stocks. His research reveals that the garment industry has high growth prospects; a view shared by many analysts. Reed identifies

Kex Ltd, a publicly listed garment trading company with sound fundamentals, as a potential investment for the fund and places a buy order for 50,000 shares. But the stock of Kex Ltd. has low liquidity. Prior to execution, Reed follows a systematic procedure of allocating shares based on the weighted value of a client's portfolio. This is a pre-defined criterion that the clients are aware of. However, when the trade is executed, Reed was able to buy only 30,000 shares. To prevent a violation of the Code and Standards, Reed allocates the shares equally among all clients. Which of the following statements is most likely correct?

A. Reed did not violate Standard III (B) Fair Dealing as he dealt fairly by distributing the 30,000 shares equally among all the client accounts.
B. Reed violated Standard III (B) Fair Dealing because a random procedure of allocating shares must be followed when the order is not fully executed.
C. Reed violated Standard III (B) Fair Dealing because he did not allocate the shares proportionately.

C is correct. Standard III (B) – Fair Dealing requires that all clients must be treated fairly when trades are allocated. In this case, a weighted allocation as per the pre-defined criteria is appropriate instead of distributing equally or randomly. Guidance for Standards I-VII

Question 9

Pierce Murdock is an investment adviser at Lancer Trust Company. He manages a significant number of individual taxable accounts and is always on the look-out for investment strategies that offer tax savings that fulfil the risk and return objectives of his clients. Murdock's clients as a result of his efforts are very happy with him and provide him with their personal and financial information. Rob Lockey, a former college class fellow of Murdock, runs his own investment management company called LIM. Lockey calls Murdock and asks him to share the financial information of his clients so that he can suggest some tax savings instruments offered by LIM. Lockey also promises to lower the sales commission charged in exchange for investment in his firm's funds. Murdock agrees to share certain relevant information since it will be in their benefit. But when asked about a certain client named Tyson, Murdock politely refuses and does not inform Lockey that he is no longer in charge of Tyson's account. Has Murdock violated the Code and Standards?

A. Yes, regarding all clients except Tyson.
B. No.
C. Yes, regarding Tyson's account only.

A is correct. Murdock has violated Standard III (E) – Preservation of Confidentiality by sharing personal information of all clients except Tyson, even though it benefits them. Murdock manages individual accounts and cannot disclose confidential information entrusted to him without his clients' permission. Guidance for Standards I-VII

Question 10

Colin Jackson, CFA, runs an investment management firm. To diversify his clients' portfolio, Jackson plans to include international investments and is in the process of hiring an external adviser to handle this asset class. His criteria for selecting the external adviser include reviewing the adviser's code of ethics, the adviser's internal control procedures, and the quality of the published return information. Jackson most likely violated Standard V (A) Diligence and Reasonable Basis by not including which of the following in his selection criteria?

A. Compliance procedures.
B. Investment process and adherence to strategy.

C. Performance measures.

B is correct. Since Standard V (A) – Diligence and Reasonable Basis mandates reviewing the investment process and adherence to its stated strategy. Guidance for Standards I-VII

Question 11

Hannah Hunt, CFA, is an independent equity analyst who posts her research reports on www.HannahHunt.com. Hunt is paid a flat fee by companies to write research reports. She then conducts a thorough research and analysis before she posts her recommendations. Hunt publishes only "buy" recommendations of stocks based on the results of her analysis. Hunt's recommendations carry weight and are much valued by the investment community. She returns the fee to the companies when her recommendations are other than "buy". Hunt does not reveal to the readers of her research reports, the fee arrangement with companies. Has Hunt violated the CFA Institute Standards in producing and distributing her equity reports?

A. Yes, by posting only "buy recommendations."
B. Yes, by accepting payments from companies for which she produces reports.
C. Yes, by not revealing the fee arrangement to her readers.

C is correct. According to Standard VI (A) – Disclosure of Conflicts, the fee arrangements must be disclosed prominently and in plain language to all her readers. Guidance for Standards I-VII

Question 12

Sam Khatri receives a call from a client, Jamison, who points out that the Richmond Fixed-Income Fund underperformed its peers at quarter-end, which is contrary to what was reported in the newsletter. The client also feels that he is not happy with the overall performance of the Richmond Fund. Khatri assures him that she will look into the matter and responds, "As a candidate in the CFA Program, I am better at valuing and managing debt securities."

Khatri researches the issue and realizes that, in fact, the Fund had not underperformed but she had missed an entry of an important trade which had led to its underperformance result. Khatri immediately informs Jamison and all other clients of this through a revised letter which indicates the missed entry and replaces it with the new performance result. She accepts the discrepancy as a human typographical error.

When referring to her candidacy in the CFA Program, did Khatri violate any CFA Standards?

A. No.
B. Yes, because she has not completed the CFA Program as yet.
C. Yes, because she improperly references the CFA designation.

C is correct. Khatri improperly references the CFA designation when she states that, "As a candidate in the CFA Program, I am better at valuing and managing debt securities." She is in violation of Standard VII (B) – Reference to CFA Institute, the CFA Designation, and the CFA Program. Guidance for Standards I-VII

Question 13

Which of the following is not a requirement of GIPS for composite construction?

A. Composites should consist of one or more portfolios.
B. The portfolios should be selected on an ex-post basis.
C. The portfolios should be managed according to a similar investment strategy.

B is correct. The determination of which portfolios to include in the composite should be done according to pre-established criteria (i.e., on an ex-ante basis), not after the fact. This prevents a firm from including only the best-performing portfolios.
A composite is an aggregation of managed according to a similar investment mandate, objective, or strategy. A composite may consist of a single portfolio when it is the only portfolio managed according to a particular mandate. *Introduction to the Global Investment Performance Standards (GIPS).*

Question 14
Mark Statham, CFA, has been working as a portfolio manager at Inwell Advisors Inc. for the past three years. The firm is updating their HR system and has asked all employees to submit their experience certificates for employment history and proof of educational qualification. Statham's employment certificates show a gap of six months when he was not employed five years back, a fact he had concealed. When probed further, he says, "I had lost my job during the 2008-09 recession. Instead of looking out for another job immediately, I took a sabbatical to strengthen my modeling skills and simultaneously prepare for Level III of the CFA exam. I passed the three levels on the first attempt. The position you were hiring me for required the CFA designation, which I had by then. I felt it was not important to highlight the unemployment period because it was so long ago and I was employed when I interviewed for this position at Inwell." Which of the following CFA Institute Standards of Professional Conduct did Statham most likely violate?

A. Standard VII (B) Reference to CFA Institute, the CFA designation, and the CFA Program.
B. None.
C. Standard I (D) Misconduct.

C is correct. As Statham knowingly concealed the period of his unemployment from his potential employer, even if it happened several years ago, his conduct reflects poorly on his professional integrity. Guidance for Standards I-VII

Question 15
Manal Ghadiali is a portfolio manager for high-net-worth individuals. While reviewing the investment performance report before it is released to clients, Ghadiali notices that the reporting system has made erroneous trade entries for two clients: Khalid Mumtaz and Zahid Malik. Correcting the trade will result in a large loss for Mumtaz and a gain for Malik. Mumtaz has earlier shown dissatisfaction with the firm. Ghadiali decides to reverse the entry, but informs the performance analyst to "hold" Mumtaz's performance report till she has a chance to follow up with the client herself. Ghadiali has most likely violated the CFA Standard related to:

A. loyalty, prudence, and care.
B. suitability.
C. market manipulation.

A is correct. Ghadiali is in violation of CFA Standard III (A) – Loyalty, Prudence, and Care. Even though the correction may lead to the client terminating his relationship with the firm, withholding information on errors would not be in the best interest of the client. Guidance for Standards I-VII

Question 16
Dan Sawyer, CFA, is an investment adviser. He has two clients, Kylie Donaherty, 35, and Matthew Connor, 33. Though the clients are similar in age, their risk profiles are starkly different. Connor is pursuing his MBA and has a family to support so he is weary of volatile investments

and wants a steady return. Donaherty, on the other hand, is single and wealthy as she has inherited a mansion and $1 million from her grandparents. She is keen on growing her portfolio by taking aggressive positions. Sawyer recommends both his clients to invest 10% of the portfolio in highly volatile, micro-cap mutual funds and 15% of the portfolio in a low liquidity small cap stock. With these recommendations, Sawyer most likely violated the CFA Institute Standards of Professional Conduct for which of the following portfolios?

A. Donaherty.
B. Connor.
C. Both the portfolios.

B is correct. The recommendations for Connor's portfolio are unsuitable as his risk tolerance is low. Guidance for Standards I-VII

Question 17

Sheila Sharma, CFA, has been employed by RYK Capital for the last 10 years, where she joined as an analyst and now is a senior portfolio manager. However, after being denied a position in the firm's investment committee, she has decided to leave RYK and start her independent practice. She is careful not to tell the firm's existing clients so as not to breach her duty to RYK by soliciting their clients. While at RYK, Sharma had developed various analytical models which she had saved on her laptop to take home on many occasions, to complete the analysis on time. Before her departure from the firm, she deletes all the information on her laptop which pertains to RYK, except the models since these were made solely by her. After leaving the firm, Sharma contacts clients of her previous firm by using public information, as there is no specific agreement with RYK which prohibits her to do so. The CFA Institute Standard most likely violated by Sharma is

A. loyalty.
B. duties to clients.
C. misconduct.

A is correct. Sharma has violated Standard IV (A) – Loyalty, by taking material with her which was RYK's property even though she had prepared the models herself. She is however free to contact her previous firm's clients as long as there is no non-compete agreement that preclude a departing employee from engaging in certain conduct. Guidance for Standards I-VII

Question 18

Brenda Lee, CFA, works as a senior relationship manager in the corporate finance department of Sinchen Securities. She has just finished reading about the upcoming expansion plans and future projections of Pioneer Materials, plc. Pioneer has selected Sinchen as one of their underwriters for a secondary offering of equity and Lee has been appointed to carry out the relevant due diligence. Lee, after analyzing the future expansion project of Pioneer, feels that the plans show only the optimistic scenario. Nevertheless, using the company's projections as the primary source, she decides to market the issue. Lee has most likely violated the CFA Standard relating to:

A. Communications with clients and prospective clients.
B. Diligence and reasonable basis.
C. Referral fees.

B is correct. Lee has most likely violated the CFA Standard V (A) relating to Diligence and Reasonable basis by using the optimistic scenario of Pioneer as the base case without analyzing the downside of the expansion, and misled the sales group – brokers about the stock price. Guidance for Standards I-VII

Question 19

Anita Dewar, CFA, is a financial analyst with AB Corporation, a conglomerate with several companies across sectors from cement to retail in India. Dewar focuses on the retail sector to identify potential companies for merger or acquisition. In one of her meetings with the CEO, she makes a case that Total Retail has a solid customer base and would be ideal for takeover. The CEO appears convinced by Dewar's argument. Over the next fortnight, she buys 5,000 shares of Total Retail for INR 500 per share, and informs the CEO about it. A month later, AB Corp. announces the company has bought out Total Retail for INR 600 a share. Which of the following standards did Dewar least likely violate by purchasing shares of Total Retail?

A. Priority of Transactions.
B. Misrepresentation.
C. Material Nonpublic Information.

B is correct. There is no evidence that Dewar has misrepresented any information. Dewar purchased the shares of Total Retail before the acquisition which is a violation of VI (B) Priority of Transactions. She also violated II (A) Material Nonpublic Information. Guidance for Standards I-VII

Question 20

Gama Corp. has hired AVI Investment Bank to underwrite its secondary public offering. AVI already has a sell recommendation on the stock which has been released by its research unit to its Brokerage Department. Which of the following actions is least appropriate to avoid violating CFA Institute Standards of Professional Conduct?

A. Place the company on a restricted list and give only factual information about Gamma.
B. AVI should discontinue issuing recommendations about Gamma.
C. Use new research to substantiate that the stock deserves a buy rating.

C is correct. A & B are consistent with Standard I (B) – Independence and Objectivity – If the firm is unwilling to permit dissemination of adverse opinions about a corporate client, members and candidates should encourage the firm to remove the controversial company from the research universe and put it on restricted list so that the firm disseminates only factual information about the company. Guidance for Standards I-VII

Question 21

New Hampshire Asset Management's current policy is to report month-to-date and year-to-date performance. New Hampshire usually publishes the performance of its Growth Stock Fund based on the last business day of the month and mentions only that the data is reported as of the last trading day of the month. However, if during the month the Fund traded at a higher level, then it mentions that data in its report without giving the exact date of the month. New Hampshire has most likely violated the CFA Institute Standard related to:

A. performance presentation.
B. preservation of confidentiality.
C. loyalty.

A is correct. New Hampshire has most likely violated Standard III (D) Performance Presentation – according to which, “When communicating investment performance, members and candidates must make reasonable efforts to ensure that it is fair, accurate and complete.” The information is not reported correctly by stating that the data is of month-end whereas at times higher traded value of the Fund is incorrectly reported as the month-end performance. *Guidance for Standards I-VII*

Question 22
Solaris Investment Company is developing an automated back office trade order receipt system to minimize time and increase efficiency of trades. Yunis Rasheed, CFA, is entrusted with overseeing the project and identifying any potential problems. Rasheed along with the compliance department reviews the preliminary developmental phase of the new system. On comparing it with the old system which is based on manual entering of trade orders at the end of day, the new system immediately recognizes a trade and issues a receipt but shows certain problems in registering small orders. It also calculates the commissions incorrectly. The new system has to meet all the regulatory requirements before the company puts it into operation. Pressed for time, Rasheed informs the programmers to correct the problems and run the necessary tests before the final execution. The programmers, accordingly correct the problems, run the required simulations and report to Rasheed that they have corrected all the issues. Soon after the new system of automated trade receipts is adopted by Solaris, Rasheed is commended by the chief operating officer (COO) for the on-time implementation of the system.
Are Rasheed’s action regarding the supervision of the new system consistent with the CFA Institute Standards?

A. Yes, because he involved the compliance department.
B. Yes, because he identified the initial problems at the developmental phase.
C. No, because he did not ascertain that the final implementation fulfilled all the regulatory requirements.

C is correct. Rasheed has violated Standard IV (C) – Responsibilities of Supervisors because he did not make sure that the final system was in accordance with the regulatory requirements. According to the Standard, Rasheed has a responsibility to make reasonable efforts to detect and prevent violations of applicable laws, rules, and regulations. As per the compliance department’s directive, Rasheed had a responsibility to ensure that the modifications corrected the potential problems, without introducing any new defects. Guidance for Standards I-VII

Question 23
Trust Williams has just been hired by Piermont investments as a research analyst. After carrying out a thorough research backed by fundamental analysis of Coneggro Foods, he rates the company as a “buy”. However, in the evening he meets a colleague, Sam Bing, from his previous firm Trust Securities, who is a reputable trader. Bing tells Williams, that Coneggro’s stock has not been doing well in the market owing to its new processed food line. The next day, Williams changes his recommendation from “buy” to “hold” and releases his report to the clients. Williams has most likely violated the CFA Institute Standard relating to:

A. Fair dealing
B. Diligence and reasonable basis.
C. Record retention.

B is correct. Williams has violated Standard V(A) – Diligence and Reasonable Basis. The opinion of another trader/friend is not an adequate basis for William’s action in changing the recommendation. A & C are incorrect since these are irrelevant in this case. Guidance for Standards I-VII

Question 24

Sophie Anthony is walking out of the hall after taking the Level I exam, and meets three of her colleagues who sat for the Level II exam. Knowing Level II was perceived as the toughest of the three levels, she asked the three candidates how they performed. The candidates said it was relatively easy and they had on an average 10 minutes to spare at the end of the exam. The candidates said they were disappointed that pension plan in FRA on which they spent a considerable amount of time did not appear on the exam. Similarly, they recall FRA formulas which were not adequately tested. Which of the following actions is most likely a violation of the CFA Institute Standards of Professional Conduct?

A. Level II candidates talking about the difficulty and length of the exam.
B. Level II candidates discussing formulas which were not tested.
C. Sophie asking the Level II candidates about the exam.

B is correct. Discussing specific topic areas and formulas that appeared on the exam is a violation of Standard VII (A) and candidate pledge. Discussing the level of difficulty or length of the exam is not a violation. Guidance for Standards I-VII

Question 25

A firm that does not adopt the GIPS standards could show misleading performance by presenting performance history:

A. considering all actual, fee-paying, discretionary portfolios managed in accordance with the same investment mandate.
B. that includes terminated portfolios.
C. for a selected time period during which the mandate produced excellent returns.

C is correct. If the firm presented performance for a selected period during which it produced excellent returns or outperformed its benchmark, such a practice would be misleading and not allowed under the GIPS standards.
A is a requirement under GIPS. On the contrary, selecting a top-performing portfolio to represent a firm's overall investment results for a specific mandate is a misleading practice that is not allowed under the GIPS standards.
B is incorrect because including terminated portfolios is consistent with the GIPS standards. Performance exclusive of such portfolios would be misleading. *Introduction to the Global Investment Performance Standards (GIPS).*

Question 26

Assume there are three parties who play different roles in the investment industry:

- Party 1: Northwest Inc. is an investment management firm that manages assets worth $500 million.
- Party 2: Aura Corp. sponsors pension plans for its employees and encourages its employees to become participants. The money is professionally managed by an investment company.
- Party 3: Futuretech Corp. is involved in the development of software for performance calculations that aids in achieving GIPS compliance.

Which of the following parties can claim GIPS compliance?

A. All the parties.
B. Only Party 1 and Party 2.
C. Only Party 1.

C is correct. Only investment management firms that manage assets can claim compliance. The other two cannot.

Question 27

Mr. Simon Brian is a Manager at a Technology Company. He invests up to USD25,000 per year in his defined contribution retirement plan. He has planned to invest USD25,000 per year in a stock index fund for the next 27 years. This fund's historical return is 12 percent per year on average. Assuming that he actually earns 12 percent a year, how much money will he have available for retirement after making the last payment?

A. $1,801,311.
B. $4,234,350.
C. $5,331,220.

B is correct.
Calculator Method
Given below are the keystrokes for computing the future value of an ordinary annuity.

Keystrokes	Explanation	Display
[2nd] [QUIT]	Return to standard calc mode	0
[2nd] [CLR TVM]	Clears TVM Worksheet	0
27 [N]	Twenty seven years	N = 27
12 [I/Y]	Set interest rate	I/Y = 12
0 [PV]	0 because there is no initial investment	PV = 0
25,000 [PMT]	Set annuity payment	PMT = 25000
[CPT] [FV]	Compute future value	FV = 4,234,350.165

Formula Method
A = USD25,000
r = 12% = 0.12
N = 27
FV annuity factor = [(1+r)N – 1] / r
FV annuity factor = [(1+12%)27 – 1] / 12% = 169.374
FVN = USD25,000(169.374) = USD4,234,350
Assuming the fund continues to earn an average of 12 percent per year, he will have USD4,234,350 available at retirement. Time value of Money. Section. 4.1. LO. e.

Question 28

The table below gives the cost structure of a company.

Output of goods	AFC	AVC	ATC
0	–	–	–

5	7.50	3.67	11.17
11	3.35	3.00	6.35
17	2.43	2.67	5.10
22	2.21	2.98	5.19
25	1.98	3.67	5.65

Based on the data above, when production increases from 17 to 22 units of goods, the marginal cost per unit of production is closest to:

A. 1.5.
B. 5.5.
C. 4.0.

B is correct. Marginal cost of production = ∆TC/∆Q.
TC at 17 = ATC x Q = 5.10 x 17 = 86.7
TC at 22 = ATC x Q = 5.19 x 22 = 114.18
For increase in output from 17 to 22, marginal cost = (114.18 – 86.7) / (22 – 17) = 5.5. Topics in Demand and Supply Analysis; Section 3.2.

Question 29

Baby World, a kids wear manufacturer, reported other operating expenses of USD25 million as on December 31, 2019. During the period under review, the accrued utilities payable increased by USD5 million while the prepaid rent expense decreased by USD7 million. Utilities and rent expense are the only two components of other operating expenses. Which of the following is most likely correct with respect to the cash paid by the Company for other operating expenses?

A. The Company paid $13 cash for other operating expenses during the period under review.
B. The Company paid $27 cash for other operating expenses during the period under review.
C. The Company paid $37 cash for other operating expenses during the period under review.

A is correct. Cash paid for other Operating Expenses = Other operating expenses – Decrease in Prepaid Expense – Increase in other accrued liabilities = USD25 – decrease in prepaid rent – increase in accrued utilities payable = USD25 – USD7 – USD5 = USD13. Understanding Cash Flow Statements; Section 3.2.

Question 30

Consider the data set:

37 25 38 43 29 35 36 24 34 30 47

Based on the above data, which one of the following is the 7th decile point?

A. 37
B. 37.4
C. 38.

B is correct. First arrange the data in ascending order and then calculate the various quantile points as follows:
24 25 29 30 34 35 36 37 38 43 47
Location of the 7th decile is:
L_{70} = (11 + 1) (70/100) = 8.4. Use linear interpolation, which estimates an unknown value based on two known values that surround it. In this case, the 8th value is 37 and the 9th value is 38. The 7th decile point, P_{70} = 37+ 0.4 (0.4 times the linear distance between 37 and 38). P_{70} = 37.4.
Organizing, Visualizing and Describing Data

Question 31
The accounting for lessors:

A. is same under IFRS and US GAAP.
B. differs based on whether the lease is classified as finance lease or an operating lease.
C. Both A and B.

C is correct. The accounting for lessors is identical under IFRS and US GAAP. However, the accounting differs based on whether the lease is a finance lease or an operating lease.
Non-Current (Long-Term) Liabilities

Question 32
Sam earns a salary of USD55,000 on his current job. He is planning to quit his job and start his own business with an initial investment of USD110,000. If he had invested this money in his friend's business, he would have earned USD63,000 per annum. Profit and loss statement for the first year of his business is given below:

Total revenue	$340,000
Cost of raw material	$30,000
Wages paid to employees	$45,000
Interest on debt	$9,000

The economic profit for Sam's business in the first year is closest to:

A. $146,000.
B. $256,000.
C. $138,000.

C is correct. Economic profit = accounting profit – total implicit costs
= 256,000 – 63,000 – 55,000
= 138,000.
The initial investment of $110,000 was not included because it is a capital expenditure and not included in profit and loss calculations.
Topics in Demand and Supply Analysis; Section 3.2.

Question 33
The following information for a company is provided as of 31 December 2015 (in $ thousands):

Cash	280
Accounts receivable	650
Inventory	2,060
Accounts payable	700
Taxes payable	420
Loan payable in three equal annual installments, each on June 30	900

Given that the industry average current ratio is 2.25, the company is:

A. **less liquid.**
B. as liquid.
C. more liquid.

A is correct. Current assets = 280 + 650 + 2060 = 2,990
Current Liabilities = 700 + 420 + (900/3) = 1420
Current ratio = Current assets ÷ Current liabilities
Current ratio = 2.11. Understanding Balance Sheets; Section 7.2.

Question 34
A mutual fund has a mean annual return of 15% and a standard deviation of return of 6%. Given the following information, the probability that the portfolio return will be between 13.5% and 18% is closest to:
$P(Z < 0.25) = 0.5987$, $P(Z < 0.50) = 0.6915$, $P(Z < 0.75) = 0.7734$, $P(Z < 1.00) = 0.8413$

A. 0.0928.
B. 0.2266.
C. **0.2902.**

C is correct.
$P(x - u / s < Z < x – u / s)$
$P(–0.25 < Z < 0.50) = P(Z < 0.50) – P(Z < –0.25) = P(Z < 0.50) – [1 – P(Z < 0.25)]$
$P(–0.25 < Z < 0.50) = 0.6915 – (1 – 0.5987) = 0.2902$. *Common Probability Distributions; Section 3.2.*

Question 35
The following table shows the GDP data of a country measured at market prices (in USD) for the year 2015:

Consumer spending on goods and services	877,560	Government spending on goods and services	654,540
Business gross fixed investment	598,730	Government gross fixed investment	99,210
Change in inventories	-87,730	Capital consumption allowance	11,230
Transfer payments	11,600	Statistical discrepancy	-5,640
Exports	512,430	Imports	530,280

The country’s gross domestic product in 2015, as per the expenditure approach, is closest to:

A. 2,787,580.
B. 2,040,100.
C. 2,118,820.

C is correct. Using the expenditures approach:
GDP = Consumer spending on goods and services + Business gross fixed investment + Change in inventories + Government spending on goods and services + Government gross fixed investment + Exports – Imports + Statistical discrepancy
GDP = 877,560 + 598,730 – 87,730 + 654,540 + 99,210 + 512,430 – 530,280 – 5,640 = 2,118,820. Aggregate Output, Prices and Economic Growth; Section 2.3.

Question 36
Capital, Inc. incurred the following costs recently on purchasing a machine for its car manufacturing plant:

Purchase price	$347,000
Delivery charges and taxes	$21,500
Installation and testing	$2,750
Floor upgradation for machine	$450
Maintenance staff training costs	$1,300

The total cost of the machine that will appear on Capital's balance sheet is closest to:

A. $368,500.
B. $371,700.
C. $373,000.

B is correct. The costs necessary for the machine to be ready to use can be capitalized. Therefore, total capitalized costs = 347,000 + 21,500 + 2,750 + 450 = USD371,700. Long Lived Assets; Section 2.1.

Question 37
Durable goods are more sensitive to business cycles as compared to non-durable goods because:

A. they have a longer useful life.
B. their purchase cannot be delayed.
C. they are needed more than non-durable goods or services.

A is correct. Durable goods are usually big-ticket items whose life span can be extended with repairs. So, consumers tend to delay replacement when the economic outlook is not favorable. Understanding Business Cycles; Section 2.2.

Question 38
The role of International Organization of Securities Commissions (IOSCO) is least likely to:

A. assist in attaining the goal of cross-border cooperation in combating violations of securities laws.
B. be responsible for regulating financial markets of member nations.
C. be the oversight body of regulatory organizations.

B is correct. IOSCO itself is not a regulatory authority technically; however, it assists in attaining this goal of uniform regulation as well as cross-border co-operation in combating violations of securities and derivatives laws. It is an umbrella organization for several major regulatory organizations over the world. Financial Reporting Quality; Section 3.3.

Question 39

Income of a country's citizens working abroad is included in:

A. **its GNP but not in its GDP.**
B. its GDP but not in its GNP.
C. both its GDP and its GNP.

A is correct. Income of a country's citizens working abroad is included in its GNP but not its GDP. International Trade and Capital Flows; Section 2.

Question 40

The confidence interval will most likely be narrowest if we:

A. **increase the sample size and reduce the degree of confidence.**
B. reduce the sample size and increase the degree of confidence.
C. increase the sample size and increase the degree of confidence.

A is correct. As we increase the degree of confidence, the confidence interval becomes wider. A larger sample size decreases the width of a confidence interval. Sampling and Estimation; Section 4.2.

Question 41

Lily Cho, equity manager, uses a stock screener and selects the following metrics: included in a global equity index, P/E ratio lower than the median P/E ratio, and a price-book value ratio lower than the median price-book value ratio. The stocks so selected would be most appropriate for portfolios of which type of investors:

A. growth investors.
B. market-oriented investors.
C. **value investors.**

C is correct. Metrics such as low P/E and low price-book are aimed at selecting value companies; therefore, the portfolio is most appropriate for value investors. Financial Statement Analysis Applications; Section 5.

Question 42

Which of the following is most likely to be the role of financial reporting?

A. **Provide information about performance, financial position and changes in financial position of a company.**
B. Report to the SEC about any potential financial fraud in a company.
C. Provide reasonable assurance about whether the financial statements as a whole are free from material misstatement.

A is correct. The role of financial reporting is to provide information about a company's performance, financial position, and changes in financial position that is useful to a wide range of users. Financial Statement Analysis; Section 2.

Question 43

Ryan Adam, an analyst at Magnus Investments, Inc., is analyzing the quarterly returns for a five-year period for two managed portfolios specializing in alternative investments. The two portfolios have similar mean, standard deviation of return and expense ratios. Adam wants to test the hypothesis that the mean quarterly return on Portfolio A equaled the mean quarterly return on Portfolio B during the five-year period. Which of the following test statistics should he most likely to use to test the hypothesis?

A. $\chi^2=(n-1)(s^2)/(\sigma_0{}^2)$

B. $t_{(n-1)}= (\bar{X}-\mu_0)/(s / \sqrt{n})$

C. $t= (\bar{d}-\mu_{d_0})/s_{\bar{d}}$

C is correct. Because the two portfolios shared essentially the same set of risk factors, their returns were not independent, so a paired comparisons test is appropriate. When we have data consisting of paired observations from samples generated by normally distributed populations with unknown variances, a t-test is based on t= (d¯-μ_d0)/s_d¯ . A is incorrect because it is a chi-square test statistic, denoted by χ2. It is used for tests concerning the variance of a single normally distributed population. B is incorrect because it is test-statistic for hypothesis tests concerning a single population mean when the population sampled has unknown variance and either of the conditions below holds:
i.the sample is large, or
ii.the sample is small but the population sampled is normally distributed, or approximately normally distributed.
Hypothesis Testing. Section 3.3. LO. i.

Question 44

A dealer gives the following quote for AED/USD: 3.5473 – 3.5489. What is the bid-offer quote in USD/AED terms?

A. 0.2819 – 0.2818.
B. 0.2818 – 0.2819.
C. 0.2818 – 0.2818.

B is correct. The USD/AED bid is the reciprocal of the AED/USD offer: 1/3.5489 = 0.2818. The USD/AED offer is the reciprocal of the AED/USD bid: 1/3.5473 = 0.2819. Note that the bid always has to be lower than the offer. Currency Exchange Rates; Section 3.1.

Question 45

According to the IFRS framework, which of the following is not a fundamental qualitative characteristic of financial statements?

A. Relevance
B. Understandability
C. Faithful representation

B is correct. Understandability is a supplementary (not a fundamental) qualitive characteristic.
According to the IFRS framework,
The two fundamental qualitative characteristics are:

- Relevance: Financial statements should be useful, both for making forecasts as well as to evaluate past forecasts. They should be timely and sufficiently detailed and important facts should not be omitted.
- Faithful representation: Information presented should be complete, neutral, and free from errors.

The four supplementary qualitative characteristics are:

- Comparability: Financial statements should be consistent over time and across firms to facilitate comparisons.
- Verifiability: Independent observers should be able to verify that information reflects true economic reality.
- Timeliness: Information should be available in a timely manner.
- Understandability: Information should be presented in a simple manner, such that even users with basic business knowledge can understand it.

Financial Reporting Standards; Section 4.

Question 46
An investor is evaluating a security whose nominal interest rate in 10.8%. The real risk-free rate in the economy is 3.5%, while the default and the maturity risk premiums are 1.8% and 1.1% respectively. If the liquidity risk premium is 2.4% for the security, the inflation premium is closest to:

A. –2.0%.
B. 2.0%.
C. 4.8%.

B is correct. Nominal interest rate = real risk-free rate of return + inflation premium + risk premiums (default, liquidity, maturity premiums). Therefore, inflation premium = 10.8% – 3.5% – 1.8% – 1.1% – 2.4% = 2.0%. Time Value of Money; Section 2.

Question 47
Under which of the following market structures do firms most likely have a well-defined supply function?

A. Oligopoly.
B. Monopoly.
C. Perfect competition.

C is correct. Under perfect competition, the supply function is well defined and is equal to the marginal cost schedule of the firm. The Firm and Market Structures; Section 2.

Question 48
Which of the following statements is least accurate regarding cash flow ratios?

A. Interest coverage ratio is calculated as an operating cash flow over interest payments.
B. Debt payment ratio measures the firm’s ability to pay debts with financing cash flows.
C. Reinvestment ratio measures the firm’s ability to acquire assets with operating cash flows.

B is correct. Debt payment ratio (CFO ÷ Cash paid for long-term debt repayment) shows the firm’s ability to pay debts with operating cash flows. Understanding Cash Flow Statements; Section 4.4.

Question 49
Anita Marshall, an analyst, is forecasting the gross profit of three companies. She uses the five-year average gross margins and forecasts sales using an internal model:

Company	**Information**
Advanics Inc.	An innovator in electronic devices and enjoys healthy margins because of its technological edge. New technologies typically replace old ones every two years in this industry.
Be-Safe Co.	Manufactures and sells medicines. Product line, demand and costs for its products have been stable throughout several years.
Celebrando LLC	Recently restructured its product offerings focusing on high margin products only.

The gross profit forecast is most reliable for:

A. Advanics Inc.
B. Be-Safe Co.
C. Celebrando LLC.

B is correct. Be-Safe Co. has been offering the same products and its demand and cost structure has been stable too. Therefore, the relationship between sales and gross profit (i.e. gross margin) should be stable and most reliable. Financial Analysis Techniques; Section 8.

Question 50
Kenneth Manufacturing Company receives orders for 14 different products from different customers. Due to resource constraint, the company can only choose to fulfill 8 orders. The total number of ways in which Kenneth Company can choose the orders is closest to:

A. 3,003.
B. 40,320.
C. 121,080,960.

A is correct. Number of ways =n!/r!(n-r)!=14!/8!(14-8)!=3,003. Alternate method: When the order of choosing r objects from a total of n objects does not matter we can use the combination formula. Therefore, number of ways = 14C8 = 3003. Probability Concepts; Section 4.2.

Question 51
To fulfill its role as a medium of exchange, money should least likely:

A. have a known value.
B. have a low value relative to its weight.
C. be easily divisible.

B is correct. To fulfill its role as a medium of exchange, money should have a high value relative to its weight. Monetary and Fiscal Policy; Section 2.1.

Question 52
Jason Long Company values land using the revaluation model. At the end of the current year, the value of the land has increased and will be adjusted on the balance sheet. For the current period, if the company prepares its financial statements in accordance with IFRS, the revaluation of the land will:

A. increase the return on assets.
B. decrease the debt to equity ratio.
C. increase the net profit margin.

B is correct. Revaluing land upwards will result in higher assets and higher equity. Therefore, it will result in a lower ROA, a lower debt to equity ratio, and an unchanged net profit margin. Long Lived Assets; Section 4.

Question 53
A country's international transactions accounts data for last year are presented below in its domestic currency:

Transaction	Amount
Exports of goods and services	3,000
Import of goods and services	3,400
Investment income payments made to foreigners	550
Investment income received from foreigners	640
Net change in assets owned abroad	550
Net change in foreign-owned assets domestically	1,050
Unilateral current transfers received	65
Unilateral current transfers paid	250
Statistical discrepancy	150

The current account balance is closest to:

A. -125.
B. -675.
C. -495.

C is correct.
Transaction Amount Totals
Export of goods and services and income receipts 3,640
Export of goods and services 3,000
Investment income received from foreigners 640
Import of goods and services and income payments -3,950
Import of goods and services -3,400
Investment income payments made to foreigners -550
Net unilateral current transfers -185
Unilateral current transfers received 65
Unilateral current transfers paid -250
Current account balance -495
International Trade and Capital Flows; Section 4.2.

Question 54
Kieron Bravo took two large samples of returns from two stock exchanges in his country, TSE and PSE. In TSE, the mean returns were 7%, with a standard deviation of 3.55%. While in PSE, the mean returns and standard deviation were 8.25% and 4.25% respectively. Given a null hypothesis of $\sigma_1^2 = \sigma_2^2$ versus an alternate hypothesis of $\sigma_1^2 \neq \sigma_2^2$, which of the following is most likely to be the test statistic?

A. 0.698.
B. 1.197.
C. 1.433.

C is correct. The test that compares the variances using two independent samples from two different populations makes use of the F-distributed t-statistic = $(\sigma_1^2)/(\sigma_2^2)$.
The smaller variance is the denominator, thus, t-stat = $[4.25]^2/[3.55]^2$ = 1.433.
Hypothesis Testing; Section 2.

Question 55
Aztec Co. incurred and capitalized €2 million of development costs during the year. These costs were fully deductible immediately for tax purposes, but the company is depreciating them over two years for financial reporting purposes. Which is the most appropriate way for an analyst to incorporate the differential tax treatment in his analysis, if the company prepares its financial statements in accordance with IFRS and the company's long history of profitability is expected to continue? The analyst should include it in:

A. liabilities when calculating the company's current ratio.
B. equity when calculating the company's return on equity ratio.
C. liabilities when calculating the company's debt-to-equity ratio.

C is correct. The different treatment for tax purposes and financial reporting purposes is a temporary difference and would create a deferred tax liability. Deferred tax liabilities should be classified as debt if they are expected to reverse with subsequent tax payments. The long history of profitability implies the company will likely be paying taxes in the following years and hence an analyst could reasonably expect the temporary difference to reverse. Under IFRS all deferred tax liabilities are non-current and therefore do not affect the current ratio. Income Taxes; Section 2.

Question 56
Carty Gigs has a defined benefit pension plan with pension liabilities of USD4 million and pension assets of USD2.5 million as on 31 December 2012. Under either IFRS or US GAAP, the reporting on the company's balance sheet would be closest to which of the following?

A. Pension assets and liabilities are not required to be shown in the balance sheet but only disclosed in footnotes.
B. USD4 million is shown as a liability and USD2.5 million as an asset.
C. USD1.5 million is reported as net pension obligation.

C is correct. The company will report a net pension obligation of USD1.5 million, equal to the pension obligation (USD4 million) less the plan assets (USD2.5 million). Non-Current Liabilities; Section 4.

Question 57
An analyst estimates a simple linear time-series model by regressing revenues of a company against time measured in years. He plots the company's observed revenues and the estimated

regression line, as shown in Exhibit 1. He also plots the residuals from this regression model, as shown in Exhibit 2.

Exhibit 1: Revenues vs. Time

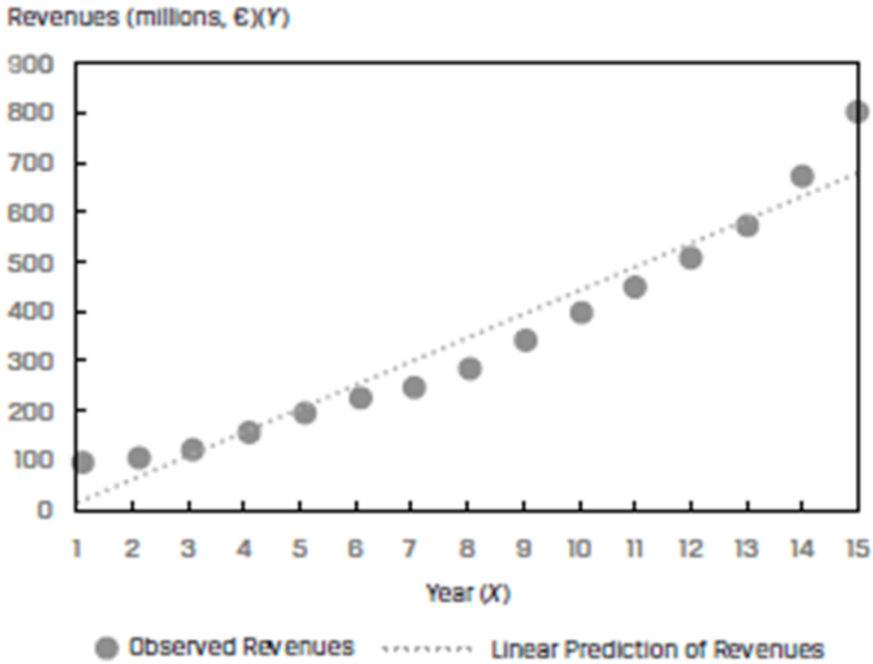
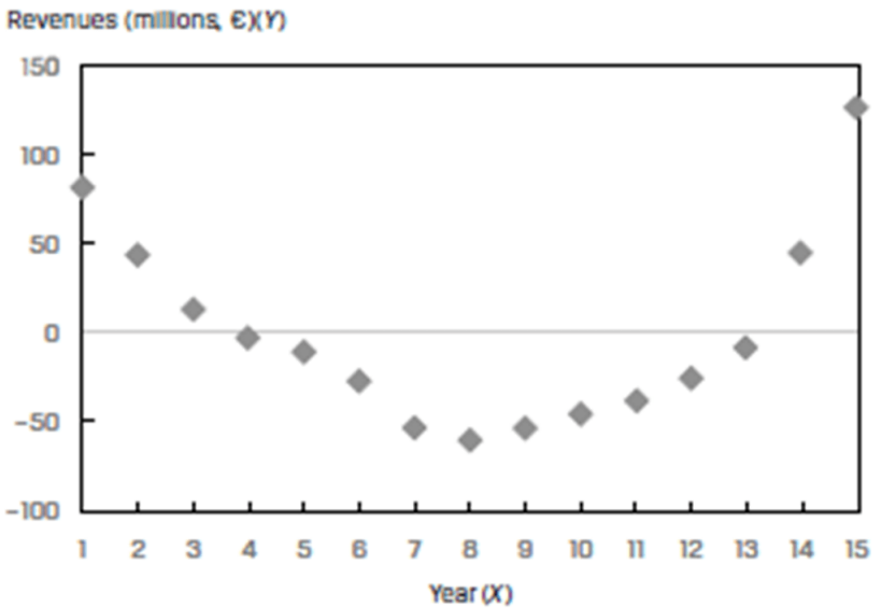


Exhibit 2: Residual Plot for Revenues vs. Time



Which of the following seems *least likely* when studying the patterns seen in Exhibits 1 and 2?

A. The correct model is non-linear as reflected in the pattern of the revenues.

B. The model is homoscedastic.

C. Lack of independence of observations could also be an issue with this model.

B is correct. The curved pattern of residuals in Exhibit 2 indicates potential heteroskedasticity i.e., the variance of the error term is not constant. If the variance of the residuals is constant for all observations, the condition is called homoskedasticity. *Introduction to Linear Regression.*

Question 58
An analyst is evaluating the relationship between the annual growth in investment (INV) in a country and the annual growth in the country's GDP (GGDP). He estimates the following two models:

	Model 1	**Model 2**
	$GGDP_i = b_0 + b_1INV_i + €_i$	$GGDP_i = b_0 + b_1 ln(INV_i) + €_i$
Intercept	1.040	1.0011
Slope	0.7739	1.883
R^2	0.615	0.945
SEE	0.517	0.231
F-stat	133.613	337.030

Based on the above data, which of the following statements in not true?

A. Model 1 is a simple linear regression with no transformations.
B. Model 2 is a lin-log model.
C. Model 1 has better goodness-of-fit.

C is correct. Model 2 has a higher R^2, higher F-stat, and a lower SEE as compared to Model 1. Therefore Model 2 has a better goodness-of-fit with the sample data. *Introduction to Linear Regression*

Question 59
Ali's monthly demand for juice is given by the equation:

$QD_{Juice}=55-1.60P_{Juice}+0.012I-1.10P_{Chips}$	
Where	QD_{Juice} is the number of bottles of juice ordered every month
	P_{Juice} is the price of a juice
	I is his monthly budget
	P_{Chips} is the price of a packet of chips

Assume that I is equal to USD50, Pjuice is equal to USD13 and Pchips is equal to USD8. The cross price elasticity of demand for juice with respect to the price of chips is closest to:

A. -0.26.
B. -0.34.
C. -0.49.

B is correct. The cross-price elasticity of demand is given by $(\Delta QD_{juice}/\Delta P_{chips})$ (P_{chips}/QD_{juice}), and notice from the demand function that $\Delta QD_{juice}/\Delta P_{chips} = -1.1$.

QD_{juice} = 55 – 1.6 (13) + 0.012 (50) – 1.1 (8) = 26
Inserting in the values for P_{chips} and QD_{juice} yields a cross-price elasticity of demand for juice of (-1.1) (8/26) = -0.34. Topics in Demand and Supply Analysis; Section 2.4.

Question 60
If a company issues USD100,000 face value zero-coupon bonds, its debt-to-equity ratio will most likely:

- **A. rise as the maturity date approaches.**
- B. decline as the maturity date approaches.
- C. remain constant throughout the bond life.

A is correct. As the discount is amortized over time, the value of the liability for zero-coupon bonds increases over time. Since E = A – L, the value of equity decreases as the liability increases. Hence the debt-to-equity ratio will increase. Financial Analysis Techniques; Section 4.4.

Question 61
A cinema operator is considering offering price reductions on weekday movie passes. At the normal price of USD60 per day, 360 customers are expected to buy passes each weekday. At a discounted price of USD50 per day, 510 customers are expected to buy passes each weekday. The marginal revenue per customer earned from offering the discounted price is closest to:

- A. USD20.
- **B. USD26.**
- C. USD35.

B is correct. MR = ΔTR ÷ ΔQ. In this case, change in total revenue per day equals USD3,900 [(510 × USD50) – (360 × USD60)], and change in units sold equals 150 (510 – 360). USD3,900 ÷ 150 = USD26. Topics in Demand and Supply Analysis; Section 3.2.

Question 62
According to the IFRS Conceptual Framework, the least likely general feature underlying the preparation of financial statements is:

- A. accrual basis.
- **B. matching.**
- C. materiality

B is correct. The IFRS Conceptual Framework specifies a number of general features underlying the preparation of financial statements, including materiality and accrual basis. Matching is not one of those general features; it is a general principle of expense recognition. Financial Reporting Standards; Section 5.

Question 63
At the beginning of the year, a company had total shareholders' equity consisting of €3,600,000 in common share capital and €2,200,000 in retained earnings. Following details are given for the year:
• Net income reported € 800,000
• Dividends paid € 270,000
• Treasury stock € 400,000
The total shareholders' equity at the end of the year is closest to:

A. **€ 5,930,000.**
B. € 6,730,000.
C. € 6,330,000.

A is correct. Shareholders' equity at the end of year = beginning shareholders' equity + net income + retained earnings – dividends – repurchase of stock = 3,600,000 + 2,200,000 + 800,000 – 270,000 – 400,000 = 5,930,000. Understanding Balance Sheets; Section 6.2.

Question 64
An analyst determines that a company has been successful in generating a profit in 8 out of last 10 years. Assume that the estimated probability will be the actual probability for the next decade. Using the binomial probability function, the probability (in %) that the company will be profitable in exactly 7 of the next 10 years is closest to:

A. **20.13.**
B. 35.60.
C. 44.74.

A is correct. Probability of being profitable (success) is 8/10 = 0.8 or 80 percent based on past 10 years record. The probability that exactly 7 years out of next 10 will be profitable:
${}_{10}C_7 \times (0.8)^7 \times (1-0.8)^{(10-7)} = 0.2013$. Common Probability Distributions; Section 2.2.

Question 65
A firm that holds a monopoly in the market earns a marginal revenue of USD25 which is equal to the marginal cost. The average total cost amounts to USD32. If the firm produces 3,500 units and sells them at USD40 per unit, it's total profit is closest to:

A. $87,500.
B. **$28,000.**
C. $140,000.

B is correct. The firm is selling at $40 and the average total cost is USD32. The firm is making a profit of USD8 per unit. This will result in a total profit of USD28,000. The Firm and Market Structures; Section 6.3.

Question 66
Frampton Corporation Ltd. follows US GAAP in preparation of its financial statements. The following information is available for the year ended 2013:

Cash received from customers	$17 million
Cash paid for income tax	$2.2 million
Cash paid for purchase of equipment	$25 million
Interest paid	$4.3 million
Cash paid for dividends	$1.7 million
Cash paid to retire long term debt	$19 million

The net cash from financing activities is closest to:

A. -$19 million.
B. **-$20.7 million.**

C. -$25 million.

B is correct. US GAAP considers only dividends paid and repayment of long-term debt as financing activities. Interest paid is an operating activity.
Cash flow from financing activities:
Cash paid for dividends USD1.7 million
Cash paid to retire long term debt USD19 million
Total USD20.7 million
Understanding Cash Flow Statements; Section 2.2.

Question 67
What is the most likely outcome if expansionary fiscal policy is combined with contractionary monetary policy?

A. Lower government spending, lower interest rates and higher aggregate output.
B. Higher government spending, higher interest rates and higher aggregate output.
C. Higher government spending, higher interest rates and lower aggregate output.

B is correct. Expansionary fiscal policy combined with contractionary monetary policy results in higher aggregate output, higher interest rates and increased government spending as a part of GDP. Monetary and Fiscal Policy; Section 4.

Question 68
An increase in which of the following items is least likely to result in a narrower confidence interval for the population mean?

A. Degrees of freedom.
B. Sample size.
C. Reliability factor.

C is correct. An increase in the reliability factor (confidence) increases the width of the confidence interval. Increasing the sample size and increasing the degrees of freedom both shrink the confidence interval. Sampling and Estimation; Section 4.2.

Question 69
A company wants to decrease its average receivable collection from its current 27 days to the industry average of 20 days by next year. Current year’s credit sales are USD400 million and it is expected to increase by USD80 million next year. The required change in the company’s average accounts receivable balance from current year to next year in order to achieve the target of decrease in collection period is closest to:

A. $ 4,383,562.
B. $ 3,287,671.
C. $ 1,534,247.

B is correct.
Current accounts receivable turnover for year 1 = 365/27 = 13.52
Target accounts receivable turnover for year 2 = 365/20 = 18.25
Accounts receivable balance for year 1 = Sales/turnover=400,000,000/13.52 = 29,589,041.
Accounts receivable balance for year 2 = Sales/turnover = 480,000,000/18.25 = 26,301,370.

The decrease of $3,287,671 is required to meet their target. Financial Analysis Techniques; Section 4.2.

Question 70
A review of a company's inventory records for the year indicates that the following costs were incurred (in USD millions):
• Fixed production overhead: 35.7
• Direct material and direct labor used: 52.4
• Storage costs incurred during production: 4.3
• Normal waste material costs: 2.1
If the company is operating at full capacity, the total capitalized inventory cost (in USD millions) was closest to:

A. 88.1.
B. 92.4.
C. 94.5.

C is correct. The total capitalized costs include fixed production costs, the direct conversion costs of material and labor, storage costs required as part of production and normal waste costs (However, abnormal waste cost is excluded) Therefore, total capitalized inventory cost is 35.7 + 52.4 + 4.3 + 2.1 = $94.5 million. Inventories; Section 2

Question 71
Ivan Garnet, a portfolio manager, analyzes the performance of a security with data for the last 60 months. The mean monthly return for the security is 1.04% with a sample standard deviation of 0.16%. Garnet wants to test if the historical monthly return is more than 1%. The null hypothesis is that the monthly return is less than or equal to 1%. If the critical value for this test is 2, which of the following is most likely the test statistic and the decision of this test?
Test Statistic Decision

A. 0.25 Do not reject H0
B. 1.94 Do not reject H0
C. 15.0 Reject H0

B is correct.
$t=(1.04-1)/(0.16/\sqrt{60})=1.94$
Since 1.94 is less than 2, the critical value, we cannot reject the null hypothesis. Hypothesis Testing; Section 2.

Question 72
A decrease in the real exchange rate quoted in terms of USD per unit of GBP is most likely associated with:

A. an increase in the USD/GBP spot exchange rate.
B. an increase in the price level in the United States.
C. an increase in the price level in Great Britain.

B is correct. The real exchange rate = spot exchange rate x price level in base currency (GBP) country / price level in price currency (USD) country. If the US price level increases the real exchange rate will fall. Currency Exchange Rates; Section 2.

Question 73
An Australian printing company which prepares its financial statements according to IFRS has experienced a decline in the demand for its products. The following information relates to the company's printing equipment as of 31 December 2013:
AUD
Carrying value of equipment (net book value) 300,000
Fair Value 280,000
Costs to sell 40,000
Value in use 250,000
The impairment loss (in AUD) is closest to:

A. 0.
B. 50,000.
C. 60,000.

B is correct. Under IFRS, an asset is considered to be impaired when its carrying amount exceeds its recoverable amount (the higher of fair value less cost to sell or value in use).
Fair value less costs to sell: 280,000 – 40,000 = 240,000
Value in use = 250,000
Recoverable amount (higher value) = 250,000
Impairment loss under IFRS = Carrying value – recoverable amount = 300,000 – 250,000 = 50,000. Long Lived Assets; Section 5.

Question 74
PowerChem bought a chemical processing machine in 2014 for £40,000. The following information is given about the machinery:

	31 December 2015	31 December 2014
Carrying amount	£25,000	£35,000
Tax base	£22,000	£34,000
Tax rate	40%	30%

What is the most likely effect on the deferred tax liability due to a change in tax rate?

A. An increase of £300.
B. An increase of £900.
C. A decrease of £1,200.

A is correct. Change in deferred tax liability = (carrying amount – tax base) × change in rate = 3,000 × 10% = £300. Income Taxes; Section 3.3.

Question 75
An analyst is analyzing performance of two equity indexes – FTSE 100 and S&P500. He wants to forecast the probability of either the FTSE 100 or the S&P500 index will fall. The probability that FTSE may rise is 45% and fall is 55% while the probability that S&P500 may rise is 65% and fall is 35%. The probability that both indexes may fall is 40%. Assuming that the two events are not mutually exclusive, which probability rule should the analyst most likely select?

A. Addition rule
B. Multiplication rule
C. Total probability rule

A is correct. Addition rule is used to determine the probability that at least one of the events will occur. It is expressed as: P(A or B) = P(A) + P(B) – P(AB). The analyst wants to forecast the probability of either the FTSE 100 or the S&P500 index will fall; hence, he will use addition rule. Multiplication rule is used to determine the joint probability of two events. It is expressed as: P(AB) = P(A|B) P(B)
The total probability rule is used to calculate the unconditional probability of an event, given conditional probabilities. It is expressed as: P(A) = P(A|S1) P(S1) + P(A|S2) P(S2) + … + P(A|Sn) P(Sn). Probability Concepts. Section 2. LO. e.

Question 76
An analyst has gathered the following information about a small country:

Account Name	**Amount (£ million)**
Consumption	78.6
Capital consumption allowance	6.4
Government spending	12.7
Imports	5.9
Gross private domestic investment	15.5
Exports	5.1

Based on the given information, the country's gross domestic product and national income (in £ million) are respectively *closest* to:

A. 112.4 and 107.8.
B. 106.0 and 99.6.
C. 93.3 and 82.3.

B is correct. GDP = Consumption + Gross private domestic investment + Government Spending + Exports – Imports = 78.6 + 15.5 + 12.7 + 5.1 – 5.9 = 106.
National income = GDP – CCA = 106 – 6.4 = 99.6. Aggregate Output, Prices and Economic Growth; Section 2.3.

Question 77
Which of the following is a necessary revenue recognition criterion under IFRS?

A. Costs can be reliably measured.
B. Goods have been delivered to the customer.
C. Payment has been received.

A is correct. The IFRS conditions that should be met include that the costs incurred can be reliably measured, and it is likely that the economic benefits will flow to the entity, not the actual receipt of any payment, and that the significant risks and rewards of ownership have been transferred, which is normally when the goods have been delivered, but not always. Understanding Income Statements; Section 3.1.

Question 78

An analyst is testing if the average returns of all stocks in the S&P 500 are greater than 10%. He has assumed a 90% confidence level. The null and alternative hypothesis is defined as:

H0: $\mu \leq 10\%$ and Ha: $\mu > 10\%$

The probability of making a Type I error is most likely:

A. 90%
B. 95%
C. 10%

C is correct. Level of significance (α) = P (Type I error)
Level of significance (α) = (1 – level of confidence) = (1 – 90%) = 10%. Hypothesis Testing. Section. 2. LO. c.

Question 79

Analyst 1: Headline inflation refers to the inflation rate calculated based on the price index of goods and services except food and energy.

Analyst 2: Core inflation usually refers to the inflation rate calculated based on a price index that includes all goods and services in an economy.

Which analyst's statement is least likely correct?

A. Analyst 1.
B. Analyst 2.
C. Both.

C is correct. Headline inflation refers to the inflation rate calculated based on the price index that includes all goods and services in an economy. Core inflation usually refers to the inflation rate calculated based on a price index of goods and services except food and energy. Understanding Business Cycles; Section 4.2.

Question 80

Galaxy Cements Research department worked on a new plant project during the year 2019. The design, construction and operation of a pilot plant has been completed but its scale is not yet economically feasible for commercial production. The following is a summary of the expenses of the research division (in millions USD):

Material and services: USD60
Direct labor: USD20
Administrative personnel: USD10
Design, construction and testing: USD365

Which of the following is most likely correct?

A. The project will be classified as in the research stage, so all costs are expensed.
B. The project will be classified as in the development stage, so costs of $445 million are capitalized.
C. The project will be classified as in the development stage, so costs of $455 million are capitalized.

B is correct. The project will be classified as in the development stage, so costs capitalized are as USD60 + USD20 + USD365 = USD445. Administrative personnel costs are always expensed. Understanding Balance Sheet. Section. 4.3.1. LO. e.

Question 81

In an inflationary environment, which of the following methods is most likely to report the highest amount of ending inventory?

A. FIFO.
B. LIFO.
C. Weighted average cost.

A is correct. In an inflationary environment, FIFO is most likely to report the highest amount of ending inventory, and LIFO is most likely to report the lowest amount of ending inventory. Inventories; Section 3.2.

Question 82

The joint probability that an investor receives USD20 for Stock A and USD13 for Stock B is 0.54, while the joint probability that she receives USD17 for Stock A and USD22 for Stock B is 0.46. The covariance of the prices of Stock A and Stock B is closest to:

A. –0.36.
B. –6.71.
C. –10.46.

B is correct.
E[X] = 0.54(20) + 0.46(17) = 18.62.
E[Y] = 0.54(13) + 0.46(22) = 17.14.
Covariance [XY] = 0.54(20 – 18.62) × (13 – 17.14) + 0.46(17 – 18.62) × (22 – 17.14) = –6.7068.
Probability Concepts; Section 3.

Question 83

The following information is available about a company:

(all figures in $ thousands)	**2011**	**2010**
Deferred tax assets	450	300
Deferred tax liabilities	(520)	(350)
Net deferred tax liabilities	(70)	(50)
Income tax payable	2300	

The company's 2011 income tax expense (in thousands) is closest to:

A. $2,300.
B. $2,320.
C. $2,520.

B is correct. Income tax expense reported on the income statement = Income tax payable + change in net deferred tax liabilities. The change in the net deferred tax liability is USD20. Therefore, the income tax expense = 2,300 + 20 = 2,320. Income Taxes; Section 2.

Question 84

A company issued USD1,000,000 of bonds with a 10-year maturity at 98. Five years later, the company called the bonds at 102 when the unamortized discount was USD10,000. The company would most likely report a loss of:

A. **$30,000.**
B. $50,000.
C. $60,000.

A is correct.
Redemption Cost = 1,000,000 x 102/100 = 1,020,000
Carrying amount retired = 1,000,000 – 10,000 = 990,000
Loss on redemption = 1,020,000 – 990,000 = 30,000. Non-Current Liabilities; Section 2.4.

Question 85
Three countries produce leather and cotton. The following table illustrates the output per worker per day in each country:

Country	Leather	Cotton
A	7	9
B	5	8
C	3	6

Which country is most likely to have an absolute advantage for producing cotton?

A. **Country A.**
B. Country B.
C. Country C.

A is correct. A country is said to have an absolute advantage in the production of a good if it can produce the good at a lower cost, in terms of resources than another country. Country A has the highest output per unit of labor. International Trade and Capital Flows; Section 2.4.

Question 86
Peter Lynch gathered the following data for a company (USD millions):

	31 Dec 11	**31 Dec 12**
Gross investment in fixed assets	$ 4.5	$ 4.5
Accumulated depreciation	$ 1.8	$ 2.7

What would be the number of years of useful life which have passed (average age) and average life of assets at installation (depreciable life) of the company's fixed assets at the end of 2012?

Average Age & Depreciable Life

A. **3 years & 5 years**
B. 2 years & 3 years
C. 2 years & 5 years

A is correct. Depreciation expense is equal to the difference in accumulated depreciation at the beginning and at the end of the year i.e. $0.9 million. Average age is equal to accumulated depreciation/ depreciation expense, or 2.7/0.9 = 3 years. Average depreciable life is equal to ending gross investment/depreciation expense = 4.5/0.9 = 5 years. Financial Reporting Quality; Section 4.2.

Question 87
Which of the following statement is most likely correct with respect to continuously compounded return?

A. Like holding period return, the continuously compounded return to time T is the sum of the one-period continuously compounded returns.
B. Like holding period return, the continuously compounded return to time T is the multiplication of one-period continuously compounded returns.
C. Unlike holding period return, the continuously compounded return to time T is the sum of the one-period continuously compounded returns.

C is correct. The continuously compounded return to time T is the sum of the one-period continuously compounded returns i.e. $r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \ldots r_{0,1}$. Using holding period returns to find the ending value of a USD1 investment involves the multiplication of quantities (1 + holding period return). Using continuously compounded returns involves addition. Common Probability Distributions; Section 3.4.

Question 88
Considering the 180-day Libor rates (annualized) for the USD and EUR are 1.685 percent and 1.642 percent respectively, and the spot EUR/USD exchange rate is 0.5498, the 180-day forward rate (FEUR/USD) is closest to:

A. 0.5469.
B. 0.5481.
C. 0.5497.

C is correct. Forward rate = Spot rate *(1 + i_{price} currency)/(1 + $i_{basecurrency}$)
Forward rate = 0.5498 *(1 + 0.01642 *180/360)/(1 + 0.01685 *180/360)
Forward rate = 0.5498 *1.00821/1.008425 = 0.5497. Currency Exchange Rates; Section 3.3.

Question 89
An analyst opts for calculating the price/tangible book value instead of the price/book value to analyze a company's financial statements that follows US GAAP. She most likely selects this ratio if the company:

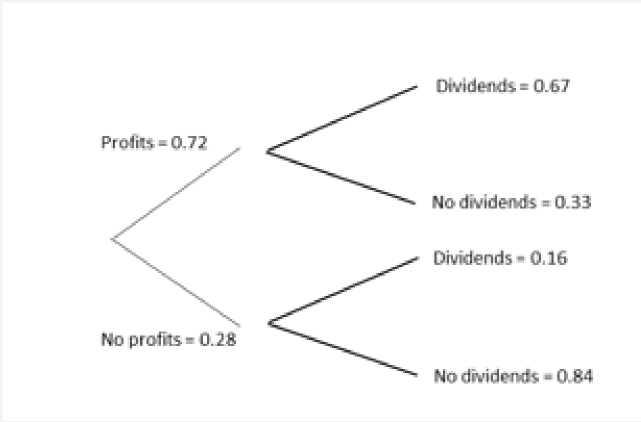
A. develops its patents and processes internally.
B. grows primarily through acquisitions.
C. invests a substantial amount in new capital assets.

B is correct. A company that grows primarily through acquisition will have more goodwill and other intangible assets on its balance sheet than an internally grown company. To compare companies with different growth strategies, an analyst would opt for price/tangible book value ratio. Financial Statement Analysis Applications; Section 6.

Question 90
The probability that a company will earn profits this year is 72%, whereas the probability of not earning profits is 28%. If the company earns profits, the probability that dividends are paid is 67%, while if the company does not earn profits, the probability of paying from company reserves is 16%. The unconditional probability that dividend is not paid is closest to:

A. 47.28%.
B. 53.72%.
C. 89.28%.

A is correct. Consider the tree diagram below:



Unconditional probability of dividend non-payment = 1 – [(0.72 × 0.67) + (0.28 × 0.16)] = 47.28%.
Probability Concepts; Section 2.